

NAME:

STUDENT ID:

These problems provide additional practice with the material from the Intro to Probability notes. Unless stated otherwise, assume that all outcomes in the stated outcome space are *equally likely*.

### Outcome spaces, events, and equally likely outcomes

1. A fair six-sided die is rolled *once*. Write the outcome space  $\Omega$  and define the event  $A$  that the result is greater than 4 by listing its outcomes. Then find  $P(A)$ .
2. A coin is tossed twice. Write the complete outcome space, being careful to distinguish the two possible orders in which one head and one tail can occur. Let  $A$  be the event that at least one toss is heads. Find  $P(A)$ .
3. A eight-sided die has sides labeled 1, 2, 3, ..., 8. Let  $A$  be the event that the result is a multiple of 3, and let  $B$  be the event that the result is odd. List  $A$ ,  $B$ ,  $A \cap B$ , and  $A \cup B$ .
4. A box contains five tickets labeled 1, 2, 2, 3, 4. One ticket is drawn at random. Let  $A$  be the event that the number drawn is *even*. Find  $P(A)$ . Why is it incorrect to treat the four distinct labels 1, 2, 3, 4 as equally likely outcomes?

### Axioms, unions, intersections, and complements

5. Let  $A$  be the event that a fair die roll is even and  $B$  be the event that a fair die roll is *greater* than 3. List  $A \cap B$  and  $A \cup B$ , and find the probability of each.
6. A pair of fair six-sided dice is rolled, and the spots are summed. Let  $A$  be the event that the sum is at least 5. Use the **complement** rule to find  $P(A)$  by first finding  $P(A^C)$ .
7. A fair coin is tossed three times. Let  $A$  be the event that all three tosses are tails. Use the complement rule to find the probability of getting at least one head. You may list the outcome space to justify your answer.

8. Suppose  $P(A) = 0.55$ ,  $P(B) = 0.40$ , and  $P(A \cap B) = 0.15$ . Find  $P(A \cup B)$ . Then find the probability that neither  $A$  nor  $B$  occurs.
9. An outcome space has events  $A$  and  $B$  with  $P(A) = 0.6$ ,  $P(B) = 0.5$ , and  $P(A \cup B) = 0.8$ . Find  $P(A \cap B)$ . Explain why simply adding  $P(A)$  and  $P(B)$  would overcount some outcomes.

## Box models and repeated draws

10. Using the same box  $\text{\texttt{H}}, \text{\texttt{H}}, \text{\texttt{T}}$ , draw twice with replacement. List the nine equally likely ticket-level outcomes and find the probability of two heads.
11. A box contains  $\text{\texttt{1}}, \text{\texttt{2}}, \text{\texttt{2}}, \text{\texttt{3}}$ . One ticket is drawn. Let  $A$  be the event that the draw is at least 2. Find  $P(A)$  and  $P(A^C)$ , and verify that they add to 1.
12. An American roulette wheel has 18 red pockets, 18 black pockets, and 2 green pockets. Find the probability of a red result, the probability of a non-red result, and the probability of winning a one-dollar net gain when betting on red.

## Probability and simulation

13. A student writes the following R code to simulate a fair die roll:

```
sample(1:6, size = 1)
```

```
[1] 4
```

Explain what the code does and why each of the six values should be equally likely.

14. Write R code using `sample()` to simulate drawing one ticket from the box  $\text{\texttt{1}}, \text{\texttt{2}}, \text{\texttt{2}}, \text{\texttt{3}}$ . Then write code using `replicate()` to simulate 1,000 draws and estimate the probability of drawing a 2.

15. The theoretical probability of rolling a 6 on a fair die is  $1/6$ . A simulation of 20 rolls produces 7 sixes. Is the simulation result impossible? Explain the difference between a theoretical probability and an empirical proportion.
16. A student wants to simulate the sum of two fair dice and writes `sample(2:12, size = 1)`. Explain why this does not give every possible sum the correct probability. Describe, in words or R code, a better simulation using two separate die rolls.
17. Use the complement rule to find the theoretical probability of seeing at least one 6 in four independent rolls of a fair die. Then describe how you would use `replicate()` to estimate this probability empirically.
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18. A fair six-sided die is rolled once. Let  $A$  be the event that the result is odd and  $B$  the event that the result is prime. List  $A$ ,  $B$ ,  $A \cap B$ , and  $A \cup B$ , and calculate the probability of each event.
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19. An event  $A$  has probability 0.37. Find  $P(A^C)$  and explain in words what the complement event  $A^C$  represents.